MIL-A-23987(DOCKS)
30 December 1963
SUPERSEDING IN PART
MIL-W-17401(DOCKS)
28 October 1952

MILITARY SPECIFICATION

A-FRAME, PONTOON TUG; PREFABRICATED, KNOCKED DOWN

1. SCOPE

1.1 This specification covers one type and size of an A-frame used on the bow of pontoon warping tug for lowering and hoisting anchors.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-T-276 - Thimbles, Rope.
FF-T-791 - Turnbuckle.
QQ-S-741 - Steel Plates, Shapes and Bars, Carbon, Structural.
RR-W-410 - Wire Rope and Strand.
WW-P-406 - Pipe, Steel (Seamless and Welded) (For Ordinary Use).

MILITARY

MIL-T-704 - Treatment and Painting of Materiel.
MIL-E-15130 - Enamel, Ship, Exterior, Alkyd, Haze Gray, No. 27, Formula No. 5H.

FSC 2020

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:39Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

DRAWING

BUREAU OF YARDS AND DOCKS

879162 - N. L. Equipment P-Series Warping Tug A-Frame Assembly and Details.

The following additional documents are required only when level A or B preservation, packaging, and packing are required.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Steel Strapping, Flat.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116 - Preservation, Methods of.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A373 - Specification for Structural Steel for Welding.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:39Z
Check the source to verify that this is the current version before use.

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The A-frame assembly shall be furnished knocked down and shall be made up of prefabricated units consisting of item numbers 1 through 18 as listed in the Bill of Materials shown on Drawing 879162.

3.2 Material. Material shall be as specified herein and as shown on the drawing. All materials shall be new and unused. Material not definitely specified shall be of standard commercial quality suitable for the intended use. Steel plates and bars shall conform to Specification QQ-S-741, type II, class 1, or to ASTM Designation A373. Steel pipe shall conform to Specification WW-P-406, weight B, class 1.

3.3 Construction. The A-frame assembly shall be constructed as shown on Drawing 879162, and shall conform to the requirements of this specification.

3.3.1 Leg and crown block units. Legs will be welded to the crown block in field assembly. The proper leg, pad eye, and collar (MK-J1) shall be furnished disassembled for welding in the field. The sheave pin shall be of commercial grade alloy steel equivalent to American Iron and Steel Institute, type 2340 or 2345, heat treated and normalized for maximum hardness. The legs shall be of 6-inch diameter steel pipe of the size and shape shown on the drawing. Plates and pins shall be of structural steel. Cotter pins shall be of corrosion-resistant steel.

3.3.2 Stays. The stays shall be furnished assembled with thimbles and cable clips. Wire rope shall be galvanized and shall conform to Specification RR-W-410, type I, class 2, construction 1. Thimbles shall be galvanized conforming to Specification FF-T-276, type III. Cable clips shall be zinc coated and shall have forged saddles. Turnbuckles shall have forged body, jaw and jaw, conforming to Specification FF-T-791, type I, form 1, class 7.

3.3.3 Pad eye and plate units. Units shall be of structural steel, fabricated and welded, of sizes and shapes shown on the drawing. Holes for pad eyes shall be smooth, round, and of proper diameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:39Z
Check the source to verify that this is the current version before use.

3.3.4 Fasteners. Bolts, nuts, and washers shall be zinc coated, and of the characteristics, dimensions, and quantities shown on the drawing.

3.4 Interchangeability. The various units and parts are to be fabricated and assembled so that units of the same type shall be interchangeable. Critical dimensions and tolerances for all units and fittings shall be as shown on the drawing, and shall be strictly observed.

3.5 Treatment and painting. All exterior and exposed surfaces of metal parts, except threaded surfaces of fasteners for field assembly, shall be cleaned, treated, and prime-coated in accordance with Specification MIL-T-704, type B, except that paint used for intermediate and finished coats shall conform to Specification MIL-E-15130, including color. Each finished coat shall be applied to a dry film thickness of not less than 1.5 mils. The finished paint application shall be smooth in appearance, continuous in color, free of dry overspray, pinholes, orange peel, sags, or other film defects. Zinc-coated surfaces shall not be painted.

3.6 Identification marking. The units shall be marked for identification in accordance with Standard MIL-STD-130.

3.7 Workmanship.

3.7.1 Steel fabrication. Steel used in the fabrication of metal parts shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Flame cutting, using a tip suitable for the thickness of the metal, may be employed instead of shearing or sawing. Re-entrant cuts shall be made in the best possible manner. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.7.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers shall be provided in accordance with good commercial practice. All bolts, nuts, and pins shall fit holes properly and connections shall be tight.

3.7.3 Welding. Manual and machine welding processes and material shall conform to applicable codes of the American Welding Society or the American Society of Mechanical Engineers, for the type of welding to be performed. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack, or intermittent welds will not be permitted. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T18:39Z
Check the source to verify that this is the current version before use.

3.7.4 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the drawing, and otherwise to the standards of good commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Inspection lot. All A-frames offered for delivery at one time shall be considered a lot for purposes of inspection.

4.2.2 Sampling for examination. A random sample of A-frames shall be selected from each lot offered to the Government in accordance with Standard MIL-STD-105 at inspection level S-2. The Acceptable Quality Level shall be 4.0 percent defective.

4.3 Examination. Each sample selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. Corrections required, following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. Examinations shall be conducted as specified in table I. Any A-frame in the sample containing one or more major defects shall be rejected. If the number of defective A-frames in any sample exceeds the acceptance number for that sample the lot represented by the sample shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:39Z
Check the source to verify that this is the current version before use.

Table I. Classification of defects in accordance with Standard MIL-STD-105

Categories	Defects
Major:	
101	Material not as specified in 3.2, and obviously defective or not suited for the purpose.
102	Construction, including welding and fabrication, not as specified in 3.3.
103	Leg pad eyes and collars not furnished disassembled, as specified in 3.3.1.
	Guard plates not welded to crown block. Sheave pins not as specified.
	Legs not fabricated of proper size and type of steel pipe.
104	Stays not furnished assembled with thimbles and cable clips as specified in 3.3.2. Wire rope, cable clips, and turnbuckles not type specified.
105	Pad eye and plate units not fabricated and welded as specified in 3.3.3.
106	Fasteners not zinc coated and of the characteristics, dimensions, and quantities specified in 3.3.4.
107	Component parts not interchangeable, and critical dimensions and tolerances not as specified in 3.4.
108	Cleaning, pretreatment, prime coating, intermediate coating, finish coating, film thickness, and general paint application not as specified in 3.5.
109	Workmanship inferior and not as specified in 3.7.
Minor:	
201	Identification marking missing, incorrect, illegible, or not as specified in 3.6.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:39Z
Check the source to verify that this is the current version before use.

4.4 Preparation for delivery inspection. The preservation, packaging, packing, and related marking of all units produced for delivery under the contract shall be inspected to verify conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The preservation and packaging shall be level A, B, or C, as specified.

5.1.1 Levels A and B. Unpainted ferrous metal surfaces including threaded surfaces and surfaces exposed by disassembly shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in, and shall be applied in accordance with Specification MIL-P-116.

5.1.2 Level C. The complete A-frame assemblies shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. The packing shall be level A, B, or C, as specified.

5.2.1 Level A. The leg and crown block units for each A-frame assembly shall be bundled in a secured lift with 1-1/4 by 0.035 inch flat steel strapping conforming to Specification QQ-S-781, type I, class B. The stays, pad eye and plate units, fasteners, and any other unassembled components shall be packed in close-fitting boxes conforming to Specification PPP-B-601, overseas type, or PPP-B-621, class 2. The contents shall be blocked and braced to prevent movement within the boxes. Strapping shall be flat steel, type I, class B.

5.2.2 Level B. The A-frame assemblies shall be packed as specified for level A, except that the boxes shall be domestic type or class, as applicable, and strapping shall be either class A or B.

5.2.3 Level C. The complete A-frame assemblies shall be packed or prepared for shipment in a manner to insure arrival at destination in satisfactory condition and which will be acceptable to carriers at lowest rates.

5.3 Marking. The bundles and shipping containers shall be marked in accordance with Standard MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:39Z
Check the source to verify that this is the currenversion before use.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).

6.2 Transportation description. The transportation description applicable to this item is:

Rail:

Forms not otherwise indexed by name, iron or steel, structural, fabricated from material 3/16 inch or thicker. Carload minimum weight 40,000 pounds.

Motor:

Forms not otherwise indexed by name, iron or steel, structural, fabricated from material 3/16 inch or thicker. Truckload minimum weight 40,000 pounds.

Custodian:

Navy - Docks

Preparing activity:

Navy - Docks

Project No. 2020-N0002Y

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:39Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION MIL-A-23987(DOCK), -Frame, Pontoon Tug; Prefabricated, Knocked Down

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

Source: https://assist.dla.mil -- Downloaded: 202640-19T18:39Z
Check the source to verify that this is the current version before use.